

Software Requirements Specification (SRS)

Project Title: FashionHub – Online Clothing Store

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed Software Requirements Specification (SRS) for the FashionHub online clothing store. This system will allow users to browse, search, and purchase clothing items online. It will also provide administrators with tools to manage products, orders, and users.

1.2 Scope

FashionHub is a web-based e-commerce application designed to sell clothing and fashion accessories. It will provide a seamless and user-friendly platform for customers to shop online while enabling efficient backend management for administrators.

1.3 Definitions, Acronyms, and Abbreviations

User: Customer using the website to browse and purchase items.

Admin: User with backend access to manage products, users, and orders.

MongoDB: NoSQL database for storing user data, product details, and orders.

Node.js: Backend runtime environment for server-side logic.

Express.js: Framework for building the server and APIs.

HTML/CSS/JavaScript: Frontend technologies for the user interface.

1.4 References

MongoDB Documentation

Express.js Documentation

Node.js Documentation

2. Overall Description

2.1 Product Perspective

FashionHub is a standalone web application accessible via web browsers. It will interact with a MongoDB database for data persistence. The backend will be developed using Node.js and Express.js, serving REST APIs consumed by the frontend built with HTML, CSS, and JavaScript.

2.2 Product Functions

- User registration and login
- Product catalog browsing with categories and search
- Product detail viewing
- Shopping cart management
- Order placement and tracking
- Admin dashboard for product, order, and user management

2.3 User Characteristics

General users with basic internet browsing skills.

Admins with basic knowledge of managing product inventory and orders.

2.4 Constraints

The system must support modern browsers.

The database must securely store sensitive information like user passwords.

2.5 Assumptions and Dependencies

Users have internet access.

Payment gateway integration is out of scope (can be added later).

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Registration and Authentication

- Users can create an account by providing their email, username, and password.
- Users can log in and log out securely.
- Passwords must be hashed before storage.

3.1.2 Product Management (Admin)

- Admin can add new products with details like name, description, price, category, and images.
- Admin can edit or delete existing products.
- Admin can view all orders placed.

3.1.3 Product Browsing

- Users can browse products by category.
- Users can search products by name or keywords.
- Product details page shows full information.

3.1.4 Shopping Cart

- Users can add products to their cart.
- Users can modify quantities or remove items.
- Cart persists during the user session.

3.1.5 Order Placement

- Users can place orders with shipping information.
- Orders are saved in the database with status.
- Users can view their past orders.

3.2 Non-Functional Requirements

3.2.1 Performance

- The application should respond within 3 seconds for any user action.
- The database queries should be optimized for faster retrieval.

3.2.2 Security

- Use HTTPS for secure data transmission.
- Protect against common web vulnerabilities (XSS, CSRF).
- Secure password storage with hashing (e.g., bcrypt).

3.2.3 Usability

- Intuitive UI design for ease of use.
- Responsive design for different screen sizes.

3.2.4 Maintainability

- Modular code structure for frontend and backend.
- Use of RESTful APIs for separation of concerns.

4. System Architecture

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js with Express.js
- Database: MongoDB
- Communication: RESTful API

5. Database Design

Collections:

- Users
- Products
- Orders
- Cart

6. Future Enhancements

- Payment gateway integration
- Advanced search filters
- User reviews and ratings
- Email notifications